Evan Yap Zhi Kai

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LinkedIn | GitHub | Portfolio

Builds AI agents and measures whether they work. Seeking AI / software internships.



Education

Singapore University of Technology and Design Singapore
BEng (Engineering Systems and Design), MSc Technology Entrepreneurship (Year 2) *Expected May 2030*
- **Coursework:** Algorithms & Object-Oriented Programming, Machine Learning, Data Modelling & Visualisation
- **1 of 6 recipients** of the SUTD Technology Entrepreneurship Programme (STEP) Scholarship

Experience

Univers Pte Ltd Aug 2026 – Present
Intern – AI Business Consultant (Aviation, Transportation, Logistics – GTM)
- Built and deployed a source-grounded ESG research agent for energy and decarbonisation consultants (Next.js, TypeScript, Vercel AI SDK, Groq): it answers only from retrieved pages and PDFs and returns cited facility briefs or VERIFIED / REFUTED / UNVERIFIED verdicts.
- Enforced grounding in code, not just the prompt: findings citing a never-retrieved source ID are dropped, responses are Zod-validated and must state limitations, and URLs to private or cloud-metadata addresses are blocked (SSRF).
- Modelled a client's HVAC energy consumption with a regression model, and scored AI and energy-management use cases on impact, telemetry availability, feasibility and effort for the Senior Director and client stakeholders.
- Placed Top 5 of 100 in the company's EnOS website revamp challenge (Sep 2026): built a 9-page interactive site (Three.js, 7-language i18n) and pitched it live to the CEO and executives.

EvanYapFitness Sep 2024 – Jan 2026
Founder
- Grew an online coaching business to 30 paying clients in total; built a Google Form and Excel intake pipeline and a Perplexity-assisted template workflow that cut programming time per client by 50%.

Projects

EGO Telegram Personal Assistant | *TypeScript, Next.js, Vercel AI SDK, Claude API, GCP Pub/Sub* Aug 2026 – Present
- Built an LLM orchestrator that routes each text, photo or voice message to 28 tools, including finance and workout sub-agents, so one message can chain actions ("log lunch and move my 3pm"); 97% routing accuracy (38/39) on my own eval set.
- Designed every agent action to be reversible: agents search before acting and ask when several rows match, deletes are soft, undo tokens persist in Sheets across serverless cold starts, and destructive actions need a 5-minute confirmation token.
- Automated expense capture from Apple Pay (iOS Shortcuts), DBS PayLah!, Grab and e-commerce receipts via Gmail Pub/Sub push, deduplicating retries and repeat receipts; parallel parsing cut sync time ~74% vs a sequential baseline.
- Secured the bot with a user allowlist, constant-time webhook secret checks, prompt-injection guardrails, financial-data redaction in logs and AES-256-GCM encryption for stored credentials.

Parallel Match – Microsoft x Premier League Hackathon | *Python, FastAPI, React, TypeScript* Oct 2026
- Built the simulator and fork engine for our team's "what if that wasn't a foul?" app on synthetic data: it replays a seeded match to a moment, applies one change and runs hundreds of paired rollouts, reporting the win-share shift with a 95% interval.
- Orchestrated the Match Desk: a Producer plans, teammates' Narrator, Law Explainer and Historian agents draft in parallel, a tool-calling Fact-Checker verifies every number and an Editor rewrites failures, with each step shown in a trace.
- Built the FastAPI backend and an 18-language React viewer (incl. right-to-left Arabic); 197 Python tests.

Systematic Trading System | *Python, pandas, SQLite, Webull OpenAPI, pytest* Sep 2026 – Present
- Separated proposal from authority in a Python trading system: strategies only propose, and deterministic code verifies, sizes and risk-checks each order (2% daily-loss stop, 5% drawdown breaker, kill switch); live trading paused since 1 Oct 2026.
- Hardened execution on the Webull SG OpenAPI with deterministic client order IDs, reconciliation of ambiguous 5xx replies instead of retries, re-armed broker-side stops and a SHA-256 hash-chained SQLite audit ledger; 263 tests in CI.
- Tested four ideas with a walk-forward harness (no-look-ahead test, deflated Sharpe with trial counting): none beat the index, and a Claude screening layer that missed its pre-set bar (win rate 43.9% vs 47.4%) was removed.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, C++, HTML/CSS
Frameworks & APIs: Next.js, React, FastAPI, REST APIs, Google Cloud Pub/Sub, Google Workspace APIs, Zod, Pydantic
AI & Data: LLM agents, tool calling, multi-agent orchestration, structured outputs, Claude API, Groq, NumPy, pandas
Engineering: pytest, GitHub Actions CI, idempotency, audit logging, serverless deployment (Vercel)

Awards & Leadership

Awards: 2nd Place, Vietnam Grand Challenge (May 2026)
Leadership: SUTD Men's Floorball Captain; SUTD House Executive Committee (Floor Events & Hostel Works Rep)